
CISCO FTD

High Availability (Active/Standby) FMC-Managed Configuration Guide

A complete lab walkthrough covering FTD Active/Standby High Availability — from network planning and dual-device registration through HA pair creation, failover link assignment, synchronization verification, and live failover testing with validated internet continuity.

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1. Introduction

High Availability (HA) in Cisco Firepower Threat Defense provides redundancy and automatic failover to ensure continuous network protection. When the active FTD fails, the standby unit automatically takes over all interface IP and MAC addresses, preserving existing sessions without interrupting traffic flow.

FTD supports Active/Standby HA mode (not Active/Active). All management, policy deployment, and synchronization are handled centrally through the Firepower Management Center (FMC).

1.1 Scope & Objectives

- Deploy two FTD appliances managed by a single FMC
- Configure Active/Standby High Availability with a dedicated failover link
- Verify HA synchronization and failover state
- Test failover by shutting down the active unit and confirming continued connectivity
- Understand virtual MAC (vMAC) behavior during failover

2. Lab Environment

2.1 Network Topology

The lab consists of two FTD appliances (FTD1 and FTD2), a centralized FMC, management and testing workstations, and two layer-2 switches. The FTDs connect to inside and outside networks through data interfaces, with a dedicated VLAN 2 link for HA failover and state synchronization.

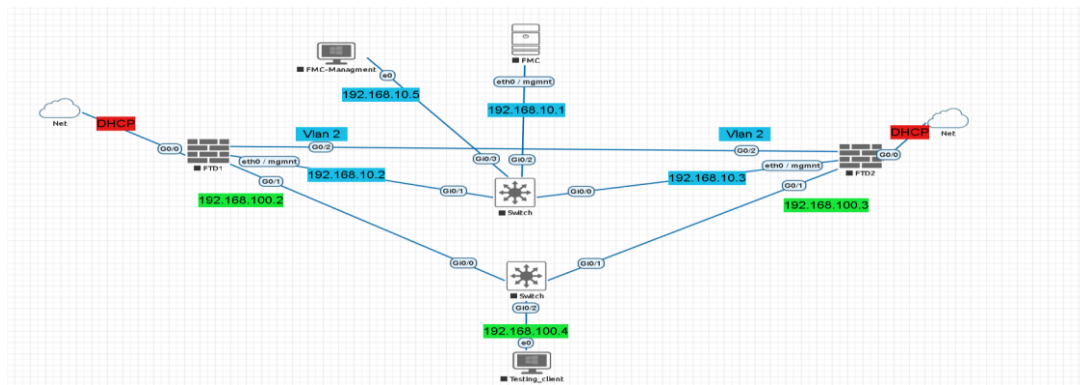


Figure 1 — HA Lab Network Topology

2.2 Management Network (192.168.10.0/24)

All management traffic (FMC-to-FTD and admin access) uses a dedicated management subnet:

Device	Interface	IP Address	Description
FMC	eth0 (mgmt)	192.168.10.1	FMC management
Management PC	e0	192.168.10.5	Admin workstation
FTD1	eth0 (mgmt)	192.168.10.2	FTD1 mgmt for FMC
FTD2	eth0 (mgmt)	192.168.10.3	FTD2 mgmt for FMC

► 2.3 Failover / State Link (VLAN 2)

The failover link carries heartbeat messages and state synchronization data between the two FTDs. It uses a dedicated data interface (GigabitEthernet0/2) on VLAN 2. FMC assigns the logical HA IPs automatically during HA pair creation — you do not manually assign IP addresses to the failover interface.

Device	Interface	VLAN	Purpose
FTD1	G0/2	2	HA Failover + State Sync
FTD2	G0/2	2	HA Failover + State Sync

► 2.4 Inside Network (192.168.100.0/24)

Device	Interface	IP Address	Description
FTD1	G0/1	192.168.100.2	Inside (Primary)
FTD2	G0/1	192.168.100.3	Inside (Secondary)
Testing Client	e0	192.168.100.4	Internal workstation
HA Virtual IP	G0/1 (shared)	192.168.100.1	FTD HA Inside Virtual IP

► 2.5 Outside Network

Both FTD outside interfaces (GigabitEthernet0/0) obtain their IP addresses via DHCP from the upstream network, simulating a typical internet-facing deployment.

3. Prerequisites

Before configuring HA, ensure all of the following requirements are met:

Requirement	Description
Same FTD Model & Version	Both FTDs must be identical in model, interfaces, and software version
Same Licenses	Both must have identical feature licenses (Threat, URL, Malware, etc.)
Same Policies	Both FTDs should have the same access control, NAT, and routing policies
Clean Configuration	Both FTDs should be freshly registered to FMC with no prior HA config
Connectivity	Management and HA link interfaces must be reachable between the two FTDs
No Pending Deployments	Deploy all changes before starting — no pending deployment leftovers

► 3.1 Interface Planning

Each FTD needs at minimum these interfaces:

Interface	Purpose
Mgmt0 (eth0)	Communication with FMC
GigabitEthernet0/0	Outside interface (internet-facing)
GigabitEthernet0/1	Inside interface (LAN-facing)
GigabitEthernet0/2	Dedicated HA Failover + State Sync link

The failover link is used to exchange heartbeat and state information (sessions, connections, NAT translations) between the two appliances. In production, consider dedicating separate interfaces for failover and state links.

4. FMC Configuration Steps

► 4.1 Step 1 — Register Both FTDs with FMC

During initial FTD setup, ensure both devices are configured for remote management (not local). Select "no" when asked to manage the device locally, and choose "routed" firewall mode:

```
> show managers
Host                : 192.168.10.10
Registration Key     : *****
Registration        : pending
RPC Status          :
```

Figure 2 — FTD initial setup: remote management, routed mode

On each FTD CLI, register with the FMC:

```
> configure manager add 192.168.10.10 key123
```

Verify the registration status shows "Pending":

```
> show managers
```

```
Manage the device locally? (yes/no) [yes]: no
Configure firewall mode? (routed/transparent) [routed]:
Configuring firewall mode ...
```

Figure 3 — configure manager output

```
> configure manager add 192.168.10.10 key123
Manager successfully configured.
Please make note of reg_key as this will be required while adding Device in FMC.
```

Figure 4 — show managers: Pending status

From the FMC GUI, navigate to Devices → Device Management → Add → Add Device. Register both FTDs individually and wait until both appear green/healthy:

The screenshot shows the FMC device registration interface. At the top, there's a 'View By:' dropdown set to 'Group'. Below it, a status bar shows 'All (1)', 'Error (0)', 'Warning (0)', 'Offline (1)', 'Normal (0)', and 'Degraded (0)'. A 'Collapse All' link is present. A table lists devices with columns: Name, Model, Ver..., Chassis, Licenses, and a status icon. One device is listed under 'Ungrouped (1)'. A right-hand panel shows a 'Register' status with a green checkmark and a 'Discovery' status with an information icon, indicating 'FTD1 - Discovery from the device is in progress.'

Name	Model	Ver...	Chassis	Licenses
FTD1 192.168.10.2	Firewall Threat Defense for...	7.3.1	N/A	Essentials

Figure 5 — FMC device registration (discovery in progress)

	Name	Model	Ver...	Chassis	Licenses	Access Control Policy	Auto RollBack
	Ungrouped (2)						
	FTD1 Short 3 192.168.10.2 - Routed	Firewall Threat Defense for...	7.3.1	N/A	Essentials	None	
	FTD2 Short 3 192.168.10.3 - Routed	Firewall Threat Defense for...	7.3.1	N/A	Essentials	None	

Figure 6 — Both FTDs registered and healthy in FMC

Configure Interfaces on Both FTDs

Before creating the HA pair, configure the data interfaces on both FTDs with matching logical names, security zones, and IP addresses. The HA-Failover interface (G0/2) should also be named consistently:

The screenshot shows the FTD1 configuration page for 'Cisco Firepower Threat Defense for KVM'. The 'Interfaces' tab is selected. Below the tabs, there's a search bar and buttons for 'Sync Interface' and 'Add Interfaces'. A table lists the configured interfaces with columns: Interface, Logical Na..., Type, Security Z..., MAC Address (Acti..., IP Address, Path M..., and Virtual Ro....

Interface	Logical Na...	Type	Security Z...	MAC Address (Acti...	IP Address	Path M...	Virtual Ro...
Diagnostic0/0	diagnostic	Phys...				Disabled	Global
GigabitEthernet0/0	Outside	Phys...	Outside		192.168.1.11/24(Sta...	Disabled	Global
GigabitEthernet0/1	Inside	Phys...	Inside		192.168.100.1/24(Sta...	Disabled	Global
GigabitEthernet0/2	HA-Failover	Phys...			192.168.250.1/30(St...	Disabled	Global
GigabitEthernet0/3		Phys...				Disabled	

Figure 7 — FTD1 interface configuration

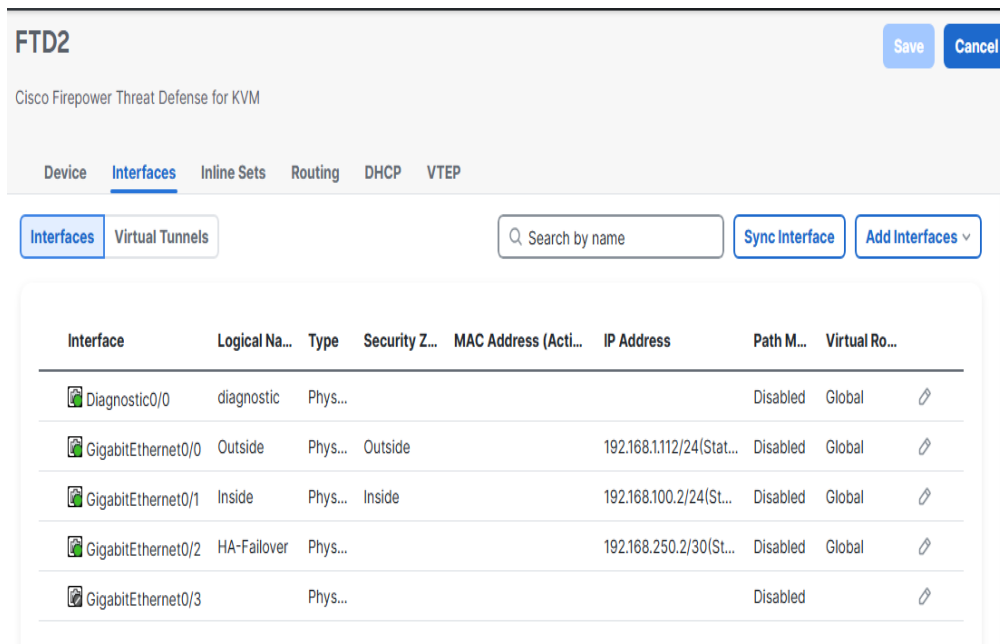


Figure 8 — FTD2 interface configuration

► 4.2 Step 2 — Create the HA Pair

With both FTDs registered and healthy, initiate the High Availability configuration:

1. In FMC, navigate to Devices → Device Management.
2. Click Add → High Availability.
3. Choose Active/Standby mode.
4. Select FTD1 as the Primary Peer and FTD2 as the Secondary Peer.
5. Click Continue.

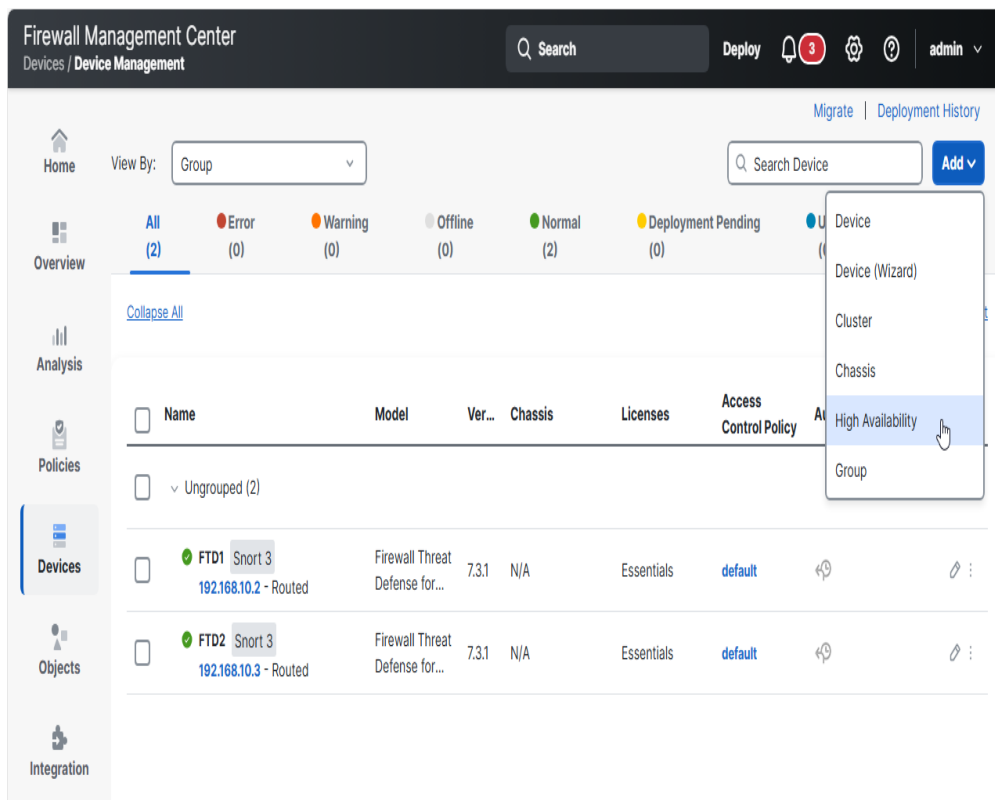


Figure 9 — FMC Device Management: Add High Availability option

Add High Availability Pair ?

Name:*

Active/passive

Primary Peer:

FTD1

Secondary Peer:

FTD2

i Threat Defense High Availability pair will have primary configuration. Licenses from primary peer will be converted to their high availability versions and applied on both peers.

Cancel

Continue

Figure 10 — Add High Availability Pair dialog

► 4.3 Step 3 — Assign Failover & State Link Interfaces

After the HA pair is created, FMC confirms the pairing and prompts you to configure interface monitoring:

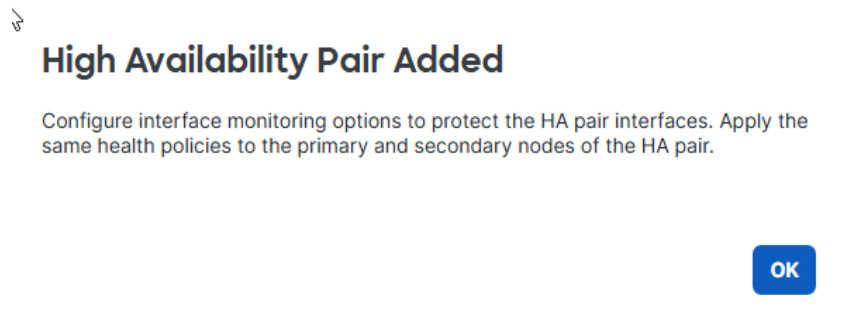


Figure 11 — High Availability Pair Added confirmation

The failover and state links serve different purposes:

- **Failover Link:** Used for control messages, heartbeats, and monitoring the peer
- **State Link:** Used to synchronize connection tables, routing tables, and NAT translations

Typically, both links share the same physical interface (GigabitEthernet0/2), though in production you may dedicate separate interfaces. FMC assigns failover IPs automatically from a /30 subnet:

FTD Role	Failover IP	State Link IP
Primary (Active)	192.168.250.1	192.168.250.1
Secondary (Standby)	192.168.250.2	192.168.250.2

Verify the failover interface assignment from the FTD CLI:

```
> show failover interface
```

```
> show failover interface
  interface Failover GigabitEthernet0/2
    System IP Address: 192.168.250.1 255.255.255.252
    My IP Address    : 192.168.250.1
    Other IP Address : 192.168.250.2
```

Figure 12 — show failover interface: IP assignments confirmed

► 4.4 Step 4 — Sync & Verify

After clicking Next, FMC will automatically:

- Verify compatibility between both FTDs (model, version, licenses)
- Synchronize configuration and policies from the primary to the secondary
- Establish the HA pair and begin heartbeat monitoring

Wait until the HA Pair Status shows: Active/Standby — Synchronized.

Verification Commands (CLI)

From either FTD CLI, use the following commands to inspect HA status:

```
> show failover
> show failover history
> show failover state
```

Expected output for a healthy HA pair:

- **This host:** Primary — Active
- **Other host:** Secondary — Standby Ready
- **Stateful Failover Logical Update Statistics:** In Sync

5. Failover Test

To validate that HA is working correctly, simulate a failure of the active FTD and confirm uninterrupted connectivity from an inside host.

► 5.1 Manual Failover (CLI)

From the active FTD, enter the diagnostic CLI and force a failover:

```
> system support diagnostic-cli
firepower# no failover active
```

The standby FTD should immediately assume the Active role. Verify by running show failover state on both units.

► 5.2 Shutdown Test

Alternatively, shut down the primary FTD entirely and observe the standby take over:

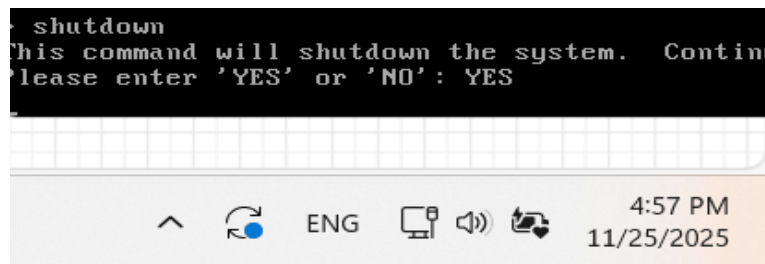


Figure 13 — Primary FTD shutdown command

From the testing client (192.168.100.4), ping 8.8.8.8 to confirm internet connectivity is maintained after failover:

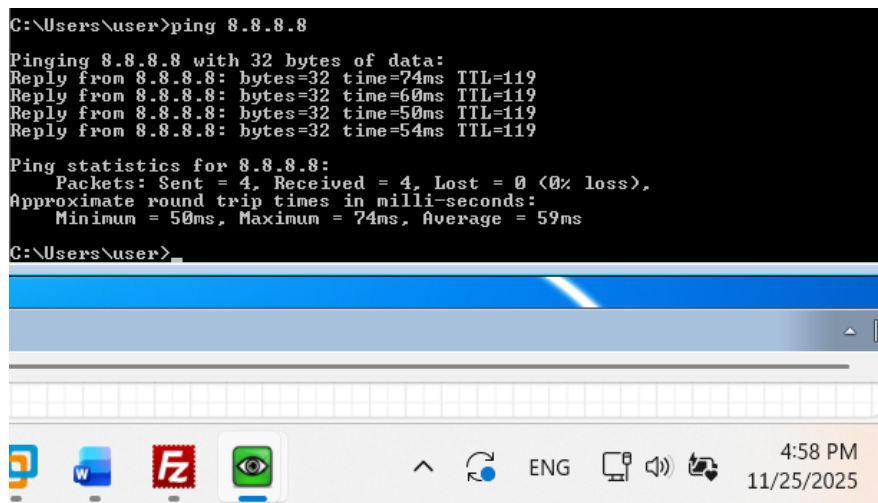


Figure 14 — Ping 8.8.8.8 successful after primary shutdown (0% loss)

The successful ping confirms that the standby FTD has taken over as active and is forwarding traffic seamlessly.

6. Active Unit Determination

The FTD uses a deterministic process to decide which unit becomes active at boot:

- If a unit boots and detects a peer already running as active, it becomes the standby unit
- If a unit boots and does not detect a peer, it becomes the active unit
- If both units boot simultaneously, the primary unit becomes active and the secondary becomes standby

7. Virtual MAC Address Concept

When two FTDs are in Active/Standby HA, only one unit forwards traffic at any time — but the network must not see any change when failover occurs. This is where Virtual MAC (vMAC) addresses are essential.

► 7.1 What is vMAC?

- A virtual MAC address is a shared MAC not tied to any single physical NIC
- The active unit uses the vMAC for traffic on each monitored interface (inside, outside)
- The standby unit knows the vMAC but does not use it until it becomes active
- FMC assigns vMAC automatically during HA configuration — no manual setup required

► 7.2 Why vMAC is Critical

Without vMAC

- When the active unit fails and standby takes over, the new active NIC has a different physical MAC
- Switches see a new MAC address → ARP caches on clients become stale
- Connections are lost until ARP tables refresh (seconds or even minutes)

With vMAC

- Switches see no MAC change during failover
- Traffic continues almost instantaneously
- TCP/UDP session tables remain intact — active connections are preserved

► 7.3 How it Works During Failover

1. Standby unit takes the Active role.
2. Activates the vMAC on the same interfaces.
3. Network sees no change in MAC addresses.
4. Sessions continue seamlessly — no ARP re-learning needed.

Think of vMAC like a team jersey: both FTDs "wear" the same jersey, but only the player on the field (active) is running. When the first player is substituted (fails), the replacement wears the same jersey — the crowd (network) notices nothing.

► 7.4 Key Behaviors

Feature	Behavior
Interface MAC seen by switch	vMAC (shared virtual address)
Active unit	Uses vMAC for all traffic
Standby unit	Does not use vMAC until it becomes Active
Failover event	No MAC change → seamless transition
Session preservation	TCP/UDP connections survive failover
Configuration	Automatic via FMC — never manually configured

8. HA Management Notes

- Only the active FTD sends logs and events to FMC
- Policy deployment happens only through FMC — it automatically syncs to both FTDs
- When the active unit fails, the standby takes over IP/MAC addresses seamlessly
- Ensure the failover link (G0/2) sits in an isolated VLAN or on isolated physical switchports — no routing, no STP interference
- If the management switch carries other traffic, use access ports or an isolated switch to avoid broadcast interference on the failover VLAN

9. Summary

Component	Details
HA Mode	Active/Standby
Managed By	Firepower Management Center (FMC)
Sync Type	Stateful (sessions, NAT, routing)
Key Interfaces	Failover Link + State Link (G0/2, VLAN 2)
Verification	show failover / show failover state
Failover Trigger	Shutdown, CLI command, or automatic on failure
Session Preservation	Yes — via vMAC and stateful sync
Typical Use Case	Redundancy for critical Firepower gateways